

# NON-DESTRUCTIVE TESTING & INTEGRITY REPORT

HYDROCARBON REFLUX UNIT · ASSET INTEGRITY DIVISION

## EQUIPMENT METADATA

### SECTION 3.2 · ULTRASONIC WALL THICKNESS MEASUREMENTS

| Casing Node                 | Nominal Baseline | Measured Thickness | Thinning Deficit | Status        |
|-----------------------------|------------------|--------------------|------------------|---------------|
| Node A-01 (Suction Nozzle)  | 4.80 mm          | 4.75 mm            | -0.05 mm         | NORMAL        |
| Node B-04 (Volute Upper)    | 4.80 mm          | 4.60 mm            | -0.20 mm         | NORMAL        |
| Node C-12 (Discharge Belly) | 4.80 mm          | 3.42 mm            | -1.38 mm         | CRITICAL WEAR |
| Node D-08 (Seal Housing)    | 4.80 mm          | 4.65 mm            | -0.15 mm         | NORMAL        |
| Node E-02 (Flange Neck)     | 5.20 mm          | 5.10 mm            | -0.10 mm         | NORMAL        |

**CRITICAL FINDING NOTE:**  
At Node C-12 (Discharge Belly Casing), ultrasonic wall thickness has degraded to 3.42 mm, representing a cumulative thinning of 1.38 mm from nominal baseline (4.80 mm). Accelerated erosion-corrosion appears localized in the lower volute quadrant. Immediate cross-referencing with OEM replacement manual is mandatory.